Adjoint-based Maximisation of Heat Flux in Models of Stellar Convection

Report Version

Callum Wright-Parish

September 8, 2026

Abstract
This report focuses on applying adjoint-based optimisation to Rayleigh-Bénard convection systems. We focus on maximising convective heat transfer in the system, measured by the Nusselt number, Nu. We use the Dedalus v3 framework, a spectral PDE solver, to simulate these systems. We aim to draw astrophysical implications by optimising the system's domain width.

Introduction

Convection is an important natural process that governs the transport of energy through many fluid dynamical systems, including stellar interiors, planetary atmospheres, and in some engineering applications. Simulating these highly turbulent astrophysical environments requires sophisticated mathematical and computational modelling. By optimising systems of Rayleigh-Bénard convection, we aim to maximise the amount of convective heat transfer within the system, parametrised by the Nusselt number, Nu, and find an appropriate scaling between this Nusselt number and the Rayleigh number, which gives a measure of the strength of thermal driving in the system.

Various papers [1][2][3][4][5] have explored methods to approximate this scaling; many use different boundary conditions or methods to obtain the maximal Nusselt number. This report aims to provide another approach to approximate this scaling law using so-called adjoint optimisation [6][7][8].

The report is structured as follows: Chapter 1 establishes the mathematical foundation by proving the adjoint-based method for both linear and non-linear cases. Chapter 2 applies this framework directly to Rayleigh-Bénard problems. We define the governing Navier-Stokes equations under the Boussinesq approximation and investigate the scaling of the Nusselt number in various steady states. This section also details the analytical setup for optimising domain width and initial conditions.

This report is a condensed summary of a broader research project. The extended report includes more detailed proofs, adjoint-based optimisation examples, and considers the impact of rotation on convection. It can be accessed on GitHub via https://github.com/callumpw06/Stellar_Convection_Research.

1 The Adjoint-Based Method

The adjoint-based method of optimisation is a highly efficient way to optimise an arbitrary number of control parameters to extremise a particular objective function under the constraints of a particular system. It is used in engineering contexts to optimise aerofoil shapes, for example [7]. We wish to apply this method to Rayleigh-Bénard systems, but before showing how it works, we need some mathematical setup.

Let us define the objective function, $J(\mathbf{y}, \mathbf{u})$, as the function we are trying to extremise with parameters:

$$\mathbf{y}(\mathbf{x}, t) = (y_1(\mathbf{x}, t), y_2(\mathbf{x}, t), ..., y_m(\mathbf{x}, t))^T \in \mathbb{R}^m \,,$$
$$\mathbf{u}(\mathbf{x}, t) = (u_1(\mathbf{x}, t), u_2(\mathbf{x}, t), ..., u_p(\mathbf{x}, t))^T \in \mathbb{R}^p \,,$$

where $\mathbf{y}$ is our state vector, and $\mathbf{u}$ is the control vector that we change. This method will find the optimal control vector that maximises J such that $\mathbf{y}$ still obeys its evolution equation.

We will use the gradient-ascent algorithm to find the optimal $\mathbf{u}$ defined by the following:

$$\mathbf{u}_{n+1} = \mathbf{u}_n + \alpha \nabla_{\mathbf{u}} J \,, \tag{1}$$

where $\nabla_{\mathbf{u}} J$ is the total functional derivative (or Fréchet derivative) of J with respect to $\mathbf{u}$, and $\alpha > 0$ is some learning rate. If we want to minimise J, we simply maximise $-J$. This algorithm requires that J is a suitably smooth function within the parameter space. This is not always the case, as it can depend on the numerical solver. In the following proof, $\mathbf{u}$ is used to define the initial conditions of the dynamical system, but this does not have to be the case (see examples in Extended Version).

1.1 Proof of the Linear Case

Let time, $t \in [0, T]$, $T > 0$, and space, $\mathbf{x} \in \Omega \subset \mathbb{R}^{m-1}$, be defined in their respective domains. Let us also define the following inner product:

$$\langle\langle \mathbf{f}, \mathbf{g} \rangle\rangle = \int_0^T \int_\Omega \mathbf{f} \cdot \mathbf{g} d\Omega dt \,.$$

For a general system involving first-order time derivatives, we define the evolution equation and initial conditions as follows:

$$\frac{\partial \mathbf{y}}{\partial t} + \mathcal{A}\mathbf{y} = \mathbf{f}(\mathbf{x}, t) \,, \tag{2}$$

$$\mathbf{y}(\mathbf{x}, 0) = \mathcal{B}\mathbf{u} \,, \tag{3}$$

where $\mathcal{A}$ is a $m \times m$ linear operator acting on $\mathbf{y}$, $\mathcal{B}$ is an $m \times p$ linear operator, and (for the sake of this proof) $\mathbf{u}(\mathbf{x})$ is part of defining the initial condition that we want to optimise. It is important to consider some assumptions we now make. We assume that $\mathbf{y}$ is defined on $\Omega \times [0, T]$, and that we can find the respective adjoint operators of $\mathcal{A}$ and $\mathcal{B}$, meaning, for some vector fields $\mathbf{f}$ and $\mathbf{g}$, there exists a linear operator, $\mathcal{A}^*$, such that

$$\langle\langle \mathbf{g}, \mathcal{A}\mathbf{h} \rangle\rangle = \langle\langle \mathcal{A}^*\mathbf{g}, \mathbf{h} \rangle\rangle \,, \tag{4}$$

and similarly, $\mathcal{B}^*$ exists. It is crucial to note that the system of PDEs that we define in (2) is linear; we will cover the non-linear case later. Now, let us construct the adjoint vector field, $\boldsymbol{\lambda}(\mathbf{x}, t) \in \mathbb{R}^m$ and some Lagrangian:

$$\mathcal{L}(\mathbf{y}, \mathbf{u}, \boldsymbol{\lambda}) = J(\mathbf{y}, \mathbf{u}) + \int_0^T \int_\Omega \boldsymbol{\lambda} \cdot \left(\mathbf{f} - \frac{\partial \mathbf{y}}{\partial t} - \mathcal{A}\mathbf{y} \right) d\Omega dt = J(\mathbf{y}, \mathbf{u}) \,. \tag{5}$$

We then use integration by parts, finding the adjoint operator, $\mathcal{A}^*$, and imposing boundary conditions that $\boldsymbol{\lambda}$ must satisfy for this to be an adjoint operator leads to

$$\begin{aligned}\mathcal{L} = J(\mathbf{y}, \mathbf{u}) &+ \int_0^T \int_\Omega \left(\frac{\partial \boldsymbol{\lambda}}{\partial t} - \mathcal{A}^*\boldsymbol{\lambda} \right) \cdot \mathbf{y} d\Omega dt + \int_0^T \int_\Omega \boldsymbol{\lambda} \cdot \mathbf{f} d\Omega dt \\ &- \int_\Omega \left(\boldsymbol{\lambda}(\mathbf{x}, T) \cdot \mathbf{y}(\mathbf{x}, T) - \boldsymbol{\lambda}(\mathbf{x}, 0) \cdot \mathcal{B}\mathbf{u}(\mathbf{x}) \right) d\Omega \,.\end{aligned}$$

By setting $\delta_{\mathbf{y}}\mathcal{L} = 0$, assuming $J(\mathbf{y},\mathbf{u}) = \int_0^T \int_\Omega F(\mathbf{y},\mathbf{u}) d\Omega dt$, and imposing boundary conditions, we find the adjoint problem

$$-\frac{\partial \lambda_i}{\partial t} + \mathcal{A}^*_{ij}\lambda_j = \frac{\partial F}{\partial y_i}\,, \quad \boldsymbol{\lambda}(\mathbf{x},T) = \mathbf{0}\,, \quad i = 1,2,...,m\,. \tag{6}$$

To find the optimal solution, we impose $\delta_{\mathbf{u}}\mathcal{L} = 0$ which eventually leads to

$$\nabla_{\mathbf{u}} J = \frac{\partial F}{\partial \mathbf{u}} + \mathcal{B}^*\boldsymbol{\lambda}(\mathbf{x},0)\,. \tag{7}$$

Thus, substituting this into (1) yields the following form of the gradient-ascent algorithm, which will form the backbone for solving our optimisation problems in this report:

$$\mathbf{u}_{n+1} = \mathbf{u}_n + \alpha\left(\frac{\partial F}{\partial \mathbf{u}} + \mathcal{B}^*\boldsymbol{\lambda}(\mathbf{x},0)\right)\,. \tag{8}$$

1.2 Non-Linear Case

If $\mathcal{A}$ is non-linear, we linearise the forward problem involving $\mathbf{y}$ to be able to find the adjoint problem. This is because we expand the variation with respect to $\mathbf{y}$ to the first order, allowing us to define the adjoint problem. If we were to expand the variation to n^{th}-order, the number of equations the system must satisfy increases by $\mathcal{O}(m^n)$. This would allow for more advanced optimisation algorithms, like Newton's method, but at great computational cost. For the gradient-descent/ascent algorithm, we only require the first-order variation of J. In practice, this method is sufficiently accurate in many cases.

1.3 The Algorithm

To carry out the adjoint-based maximisation algorithm [8] for the general use-case of $\mathbf{u}_n$, we need to carry out the following steps:

1. Start with an initial guess, $\mathbf{u}_0(\mathbf{x})$, for the optimal control vector.
2. Forward solve the system (2) using $\mathbf{u}_n(\mathbf{x})$ to define an updated problem, then find $\mathbf{y}_n(\mathbf{x},t)$.
3. Evaluate $J(\mathbf{y}_n,\mathbf{u}_n)$. If $|J(\mathbf{y}_n,\mathbf{u}_n) - J(\mathbf{y}_{n-1},\mathbf{u}_{n-1})| < \epsilon$, some tolerance, then stop.
4. Otherwise, solve the adjoint system (6) going backwards in time (starting with $t = T$) to find the adjoint field $\boldsymbol{\lambda}(\mathbf{x},t)$. In practice, we let $\tau = T - t$ and solve for $\tau \in [0,T]$.
5. Then use the adjoint field's 'final' state of $\boldsymbol{\lambda}(\mathbf{x}, t = 0)$ to update $\mathbf{u}_n$.
6. Repeat steps 2-5 until J converges, or for N iterations so $n = 1,2,...,N$.

At this stage, it is important to emphasise that these systems are rarely solved analytically, and we are focused on their numerical application. Although this method may seem complicated and require a lot of mathematical setup, it is orders of magnitude faster and more efficient than a brute-force approach. Our domain is formed of k discrete points; the brute-force method would require kN passes (a pass is a single solve of a dynamical system), whereas using the adjoint-based method simply requires $2N$ passes over N iterations. Both the forward and adjoint systems are solved numerically using Dedalus v3 [9][8], finding a balance between speed and accuracy.

2 Application to a Rayleigh-Bénard Problem

We have proven that the adjoint-based method is a viable option to optimise PDE systems in general; now we can apply it to Rayleigh-Bénard systems. We will use the widely accepted Boussinesq approximation to the compressible Navier-Stokes equations [2]. It treats changes in the fluid's density as effectively negligible - essentially constant density. That is,

$$\frac{\partial \rho}{\partial t} = 0 \implies \nabla \cdot \mathbf{v} = 0\,. \tag{9}$$

This then makes the rest of our equations much more manageable;

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} = -\frac{1}{\rho}\nabla p + \frac{\mu}{\rho}\nabla^2 \mathbf{v} + \frac{1}{\rho}\mathbf{F}\,, \tag{10}$$
$$\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T = \kappa \nabla^2 T + \Phi\,, \tag{11}$$

where Φ is some internal heating, κ is the thermal diffusivity, fluid density $\rho(\mathbf{x}, t)$, fluid velocity $\mathbf{v}(\mathbf{x}, t)$, temperature $T(\mathbf{x}, t)$, pressure $p(\mathbf{x}, t)$, and $\mathbf{F}$ representing external forces. In our problem, we will be setting $\Phi = 0$, as we focus on heating via the boundaries. We are now considering the incompressible Navier-Stokes equations, with $\rho = \rho_0$.

We non-dimensionalise our equations by scaling each variable: distance by the height of the layer of fluid, H; time by the thermal diffusion time, H^2/κ; and pressure by $\rho_0 H^2/\kappa$, as demonstrated in Goluskin, 2015 [2]. We then also use an approximation to the force created through buoyancy,

$$\frac{1}{\rho_0}\mathbf{F} = -g(1 - \beta(T - T_0))\hat{\mathbf{e}}_z\,,$$

where T_0 is the temperature of the top boundary, and β is the coefficient of thermal expansion. When we apply these to the above equations, we get the non-dimensionalised Boussinesq approximation equations:

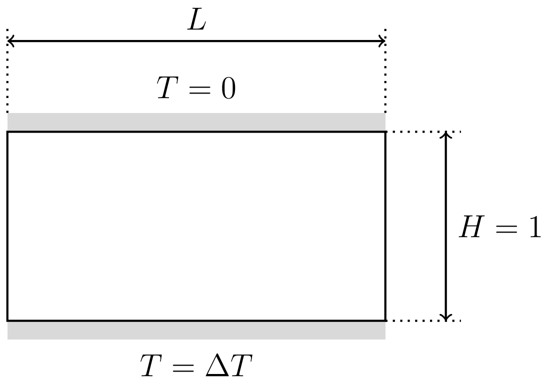
$$\nabla \cdot \mathbf{v} = 0\,, \tag{12}$$
$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} = -\nabla p + Pr \nabla^2 \mathbf{v} + Pr Ra T \hat{\mathbf{e}}_z\,, \tag{13}$$
$$\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T = \nabla^2 T + Q\,, \tag{14}$$

where Q is some internal heating that we will set to zero. We define the Rayleigh and Prandtl numbers:

$$Ra := \frac{g \beta H^3 \Delta T}{\kappa \nu}\,,$$
$$Pr := \frac{\nu}{\kappa}\,,$$

where ν is kinematic viscosity, and ΔT is the difference in temperature between the two boundaries. The Rayleigh number is a measure of how much we are thermally driving the system. There is a critical Rayleigh number, Ra_c, required for convection to occur under different boundary conditions [10][11][12][13]. The Prandtl number is a material property of each fluid, and it is a measure of how quickly a fluid diffuses momentum compared to how fast it diffuses heat.

We will be focusing on a two-dimensional (x, z), problem modelling a layer of fluid in a domain of dimensionless height $H = 1$ and width L. The fluid will be heated from the bottom and cooled at the top. Before applying the adjoint-based optimisation to this problem, we will build some further intuition by looking at the system's steady (equilibrium) states.



2.1 Steady States

We define the Nusselt number as the ratio of heat transported via convection to that transferred by conduction; in our system [2], it is calculated as follows:

$$Nu = 1 + \langle v_z T \rangle \,,$$

where v_z is the vertical component of a particles velocity, and $\langle \cdot \rangle$ represents an average across the domain [2].

To find the Nusselt number of the system at a given Rayleigh number, we run the time-dependent simulation until the system reaches its steady state: $\frac{\partial T}{\partial t} = 0$, $\frac{\partial \mathbf{v}}{\partial t} = \mathbf{0}$. However, a strictly steady state is not always reached at higher Rayleigh numbers as the system becomes more turbulent; in these cases the final state may only be statistically steady.

This steady-state condition drastically simplifies the Navier-Stokes equations, removing time dependence. For low Rayleigh numbers near the critical onset of convection ($Ra_c \approx 657.511$ for free-slip boundaries [2]), analytical approximations like the Galerkin truncation yield a Nusselt number scaling of $Nu \approx 1 + 2(1 - Ra_c/Ra)$. However, as the Rayleigh number increases, the system becomes highly turbulent, and analytical approximations become inaccurate. To predict fluid behaviour at these higher, astrophysically relevant Rayleigh numbers, we must simulate the fluid and extrapolate Nu using an appropriate scaling law.

2.2 Optimising Domain Width

At larger Rayleigh numbers, the width of the domain, L, critically dictates the fluid's convective efficiency. The spectral solver restricts the fluid to discrete Fourier modes, with wavenumbers $k_n = 2\pi n/L$ where $n \in \mathbb{Z}$. If the domain width cannot fit the natural optimal wavelength, the fluid is forced into non-optimal harmonic states, becoming less efficient. Therefore, we apply the adjoint-based method to find the optimal domain width, L^*, that strictly maximises the Nusselt number of the system's steady state, represented by $(\overline{\mathbf{v}}, \overline{T}, \overline{p})$. We define our objective function as the spatial average of the convective heat flux:

$$J = \int_0^1 \int_0^L \frac{1}{L}(1 + v_z T)\, dx dz \,.$$

To evaluate the gradient $\frac{dJ}{dL}$ without a computationally prohibitive brute-force parameter sweep, we transform the x-coordinates ($\hat{x} = x/L$) to shift the domain dependence from the integral boundaries directly into the governing operators (∇_L). By taking the first-order variation of a constructed Lagrangian with respect to the state variables, we derive the linearised adjoint problem:

$$-\overline{\mathbf{v}} \cdot \nabla_L \boldsymbol{\lambda} + (\nabla_L \overline{\mathbf{v}})^T \boldsymbol{\lambda} = \nabla_L \pi + Pr \nabla_L^2 \boldsymbol{\lambda} - \theta \nabla_L \overline{T} + \overline{T} \hat{e}_z \,,$$

$$\nabla_L \cdot \boldsymbol{\lambda} = 0 \,,$$

$$-\overline{\mathbf{v}} \cdot \nabla_L \theta = \nabla_L^2 \theta + Pr Ra \lambda_z + \overline{v_z} \,.$$

Solving this system backwards in time provides the adjoint fields $(\boldsymbol{\lambda}, \theta, \pi)$, which are then used to analytically compute $\frac{dJ}{dL}$ and update L via the gradient ascent algorithm.

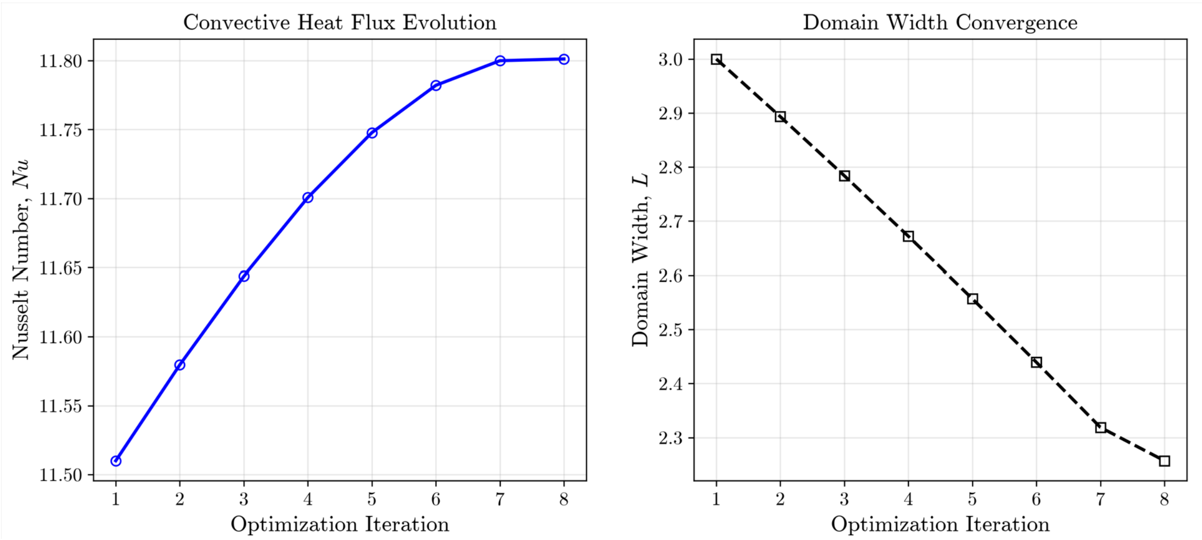


Figure 1: Adjoint optimisation converging to the optimal domain width for $Ra = 10^5$, $Pr = 1$.

As shown in the optimisation results in Figure 1, stretching the domain to its optimal width perfectly aligns the available discrete Fourier modes with the fluid's natural convective rolls. This maximises the contribution from the fundamental mode as well as the heat transport.

2.3 Asymptotic Scaling Law and Viscous Dissipation

By iteratively applying this domain-optimisation across a wide sweep of Rayleigh numbers, we can extract the fundamental empirical scaling. The maximum achievable Nusselt number for free-slip boundaries follows a remarkably clean fractional scaling law:

$$Nu \sim 1 + \frac{1}{7} Ra^{3/8} \,.$$

This result is supported by a previous report [14] that achieves a similar scaling by focusing on the fundamental modes of the steady-state solutions. We find in the extended report that the optimal solutions at these Rayleigh numbers maximise the contribution from these fundamental modes.

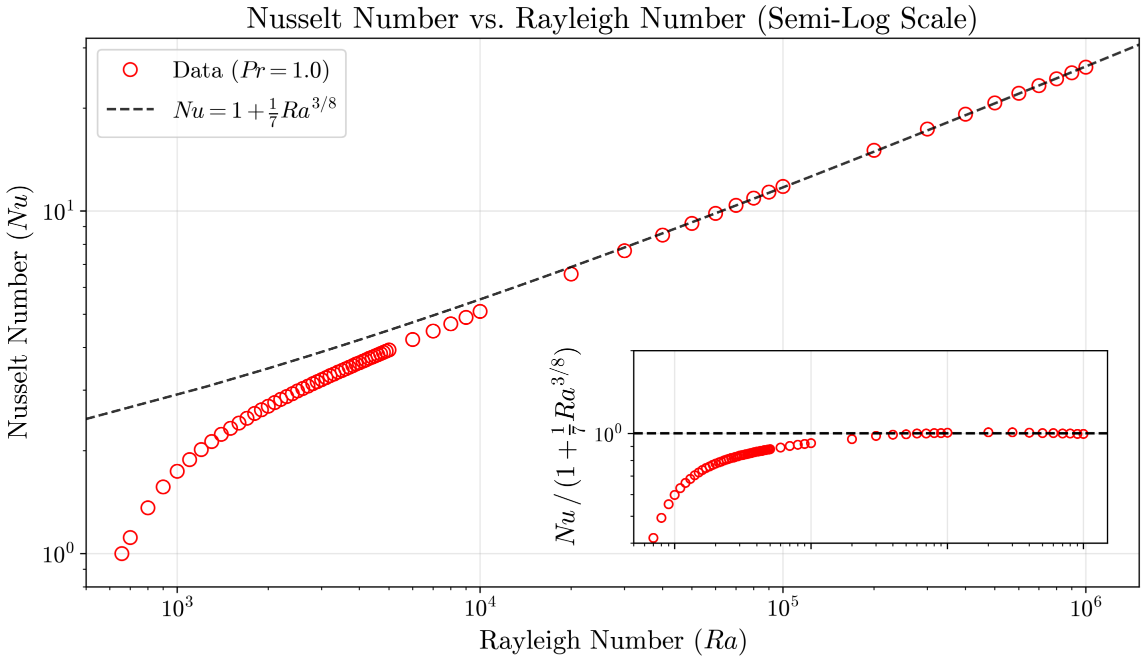


Figure 2: Optimal Nusselt number against Rayleigh number under free-slip boundary conditions.

The Nusselt number can be a measure of the potential energy of the fluid. To maintain a steady state, this energy must be expended on overcoming internal friction, measured globally as viscous dissipation, ϵ. Under free-slip conditions, our simulations confirm this exact energy balance [3]:

$$\epsilon = \sqrt{RaPr}(Nu - 1)\,.$$

Consequently, geometrically optimising the domain to maximise convective heat transfer inherently forces the fluid to maximise its global viscous dissipation. Substituting our approximate scaling for $Ra > 10^4$ into the energy balance yields:

$$\epsilon \sim \frac{1}{7}Ra^{7/8}Pr^{1/2}\,.$$

2.4 Optimising Initial Conditions

While the domain width governs the final steady state, we can also use the adjoint framework to optimise the initial temperature and velocity perturbations.

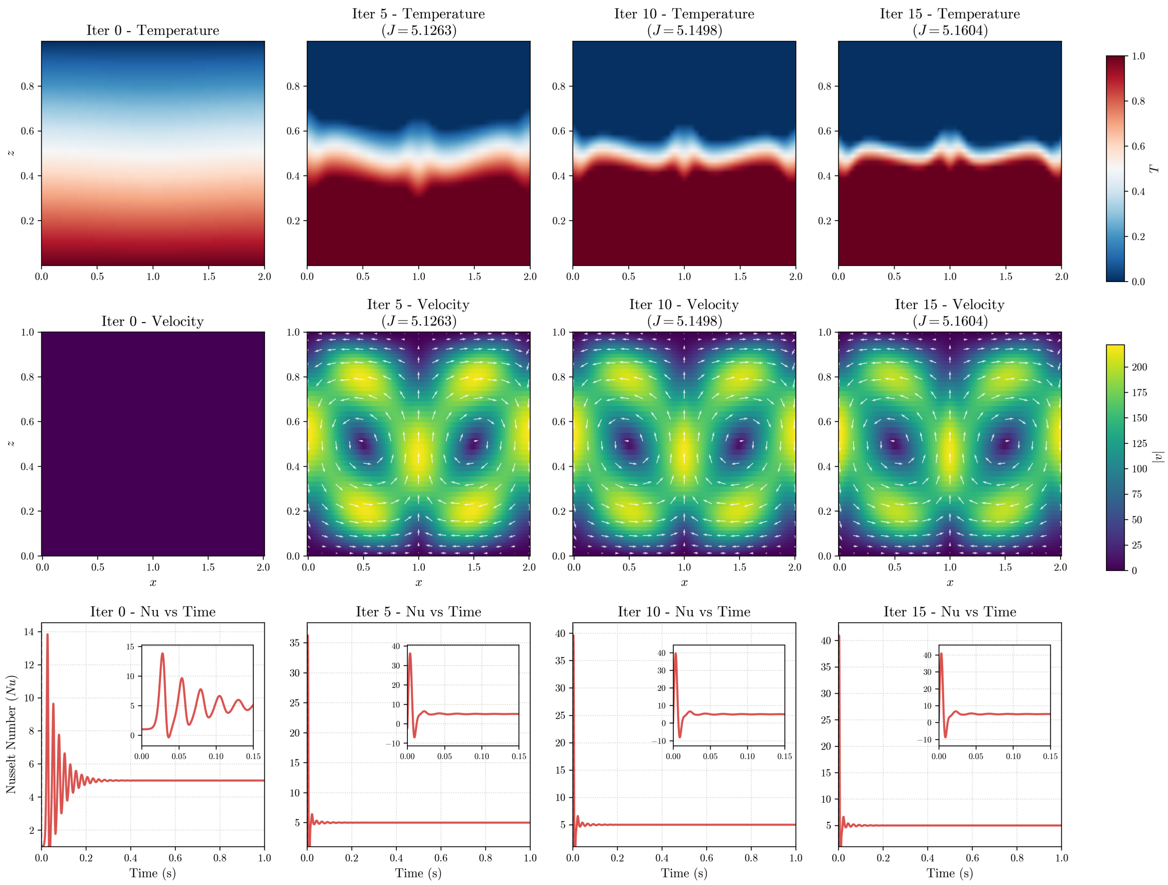


Figure 3: Evolution of initial conditions designed to maximise the time-averaged Nusselt number under no-slip boundary conditions ($Ra = 10^5$, $Pr = 1$).

Our results reveal that the optimal initial conditions successfully eliminate early-stage convective fluctuations, drastically reducing the transient time required for the system to reach its stable steady state. However, regardless of these initial conditions, the fluid's long-term asymptotic state remains identical. The initial configuration purely dictates the transient path, while the domain width dictates the final physical equilibrium.

3 Conclusions

In this report, we successfully demonstrated the power of adjoint-based optimisation to maximise convective heat transfer within Rayleigh-Bénard systems rigorously. By shifting away from traditional brute-force parameter sweeps and leveraging the continuous adjoint equations alongside a spectral solver, we established a highly efficient computational framework for discovering optimal fluid states.

By optimising the spatial domain width, we eliminated turbulent contributions from higher-order Fourier modes, maximising global viscous dissipation. Finding this optimal state allowed us to isolate a precise, theoretical asymptotic scaling law for free-slip boundaries, $Nu \sim 1 + \frac{1}{7}Ra^{3/8}$. While work still needs to be done to fully place this scaling in the context of the vast literature, this relationship offers a powerful predictive tool for extrapolating the behaviour of stellar media in extreme high-Rayleigh regimes that currently sit beyond the limits of practical direct numerical simulation.

In this project, we demonstrated our ideas using the paradigm problem of Rayleigh–Bénard convection. Natural next steps would be to incorporate the effects of rotation, magnetic fields, and fluid compressibility as we move toward more realistic models of stellar interiors.

References

[1] Sondak D. Smith L.M. Waleffe F. Optimal heat transport solutions for rayleigh-benard convection. J. Fluid Mech. ,vol.784, pp.565–595, 2015.

[2] Goluskin D. Internally heated convection and rayleigh-bénard convection. physics.flu-dyn, 2015.

[3] Grossmann S. Lohse D. Scaling in thermal convection: A unifying view. Journal of Fluid Mechanics, 2000.

[4] Alben S. Wang X. Vuong N. Enhancing wall-to-wall heat transport with unsteady flow perturbations. J. Fluid Mech. vol. 1024, 2025.

[5] Hadjerci G. Motoki S. Kawahara G. Steady solutions for the analysis of scaling regimes in rapidly rotating rayleigh-benard convection. Journal of Physics: Conference Series 3230, 2026.

[6] Giles M.B. Pierce N.A. An introduction to the adjoint approach to design. Flow, Turbulance and Combustion 65: 393-415, 2000.

[7] Schramm M. Stoevesandt B. Peinke J. Optimization of airfoils using the adjoint approach and the influence of adjoint turbulent viscosity. Special Issue Computational Methods in Wind Engineering, 2018.

[8] Skene C.S. Burns K.J. Fast automated adjoints for spectral pde solvers. PNAS Vol. 123 No. 15, 2026.

[9] Burns K. Vasil G. Oishi J. Lecoanet D. Brown B. Dedalus: A flexible framework for numerical simulations with spectral methods. Phys. Rev. Research 2, 023068, 2020.

[10] Lord Rayleigh. On convection currents in a horizontal layer of fluid, when the higher temperature is on the under side. Phil. Mag., 1916.

[11] Jeffreys H. Some cases of instability in fluid motion. Proc. R. Soc. A, 1928.

[12] Low A.R. On the criterion for stability of a layer of viscous fluid heated from below. Proc. R. Soc. A, 1929.

[13] Southwell R.V. Pellew A. On maintained convective motion in a fluid heated from below. Proc. R. Soc. A, 1940.

[14] Howard LN. Heat transport by turbulent convection. Journal of Fluid Mechanics, 1963.